

School of Mathematical Sciences
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Kevin Allen

Personal

Place of birth Dublin, Ireland
Citizenship Irish

Research Interests

Number Theory, Combinatorics, Knot Theory, Modular Forms, Projective Geometry, Cryptography

Education

- Ph.D. **Mathematics**, *University College Dublin*, 2025,
Topic: *q-Series and Modularity*.
Advisor: Professor Robert Osburn
- M.Sc. **Mathematical Science**, *University College Dublin*, 2021,
Thesis: *Semifields, Hyperovals and Cherowitzo's Conjecture*.
Advisor: Dr. John Sheekey
- B.Sc. **Mathematics**, *University College Dublin*, 2020,
Thesis: *Semifields: A Classification Problem*.
UCEAP, University of California Los Angeles, 2019

Employment

- 9/2025- **Lecturer**, *University College Cork*, Cork, Ireland.
- 9/2021 - 5/2025 **Tutor**, *University College Dublin*, Dublin, Ireland.
- 9/2019 - 5/2020 **Tutor**, *University College Dublin*, Dublin, Ireland.

Grants/Awards

- 2021-25 **Research Demonstratorship Scholarship**, *University College Dublin*, €77,400.
- 2020 **Casey Medal (awarded to top of undergraduate class)**, *University College Dublin*.
- 2018 **B.Sc. Mathematics Stage 3 Scholarship**, *University College Dublin*, €1,000.
- 2016 **UCD Entrance Scholar**, *University College Dublin*.

Publications

5. *Tails of Generalized Ranks* (with J. Maglione), in preparation.
4. *Integer partitions, primes, and integer factorization* (with W. Craig), in preparation.
3. *Generalized rank deviations for overpartitions* (with R. Osburn and M. Storzer), Accepted for publication in Illinois Journal of Mathematics.
2. *Unimodal sequences and mixed false theta functions* (with R. Osburn), Advances in Mathematics, **473** (2025), Paper No. 110293.
1. *On translation hyperovals in semifield planes* (with J. Sheekey), Des. Codes Cryptogr. **93** (2025), no. 2, 429–442.

Talks Given

- 11/2021 **Speaker at UCD Mathematical Society & UCD PhD Society**,
“Knot Theory and Untangling the Path Ahead”.
- 6/2022 **Short Talks at Nesin Mathematics Village, Workshop on Integer Partitions**,
“Knot Theory and q-Series”.
- 5/2023 **35th Automorphic Forms Workshop**,
“Unimodal sequences, Hecke-type double sums and false theta series”.
- 7/2023 **32èmes Journées Arithmétiques**,
“Unimodal sequences, Hecke-type double sums and false theta series”.
- 6/2024 **Canadian Number Theory Association Meeting**,
“Unimodal sequences and mixed modularity”.
- 1/2026 **UCC TY Work Experience Week**,
“Number Bases”.
- 3/2026 **University of Galway, School of Math. & Stat. Sciences Seminar**,
“Partitions, Overpartitions and Rank Deviations”.

Conferences Attended

- 5/2022 **NSF/CBMS Research Conference: Ramanujan’s ranks, Mock Theta Functions, and Beyond**, University of Texas Rio Grande Valley, Texas, USA, May 16-20, 2022.
- 6/2022 **Workshop on Integer Partitions**, Nesin Mathematics Village, Turkey, May 30-June 5, 2022.
- 7/2022 **Ramanujan and Euler: Partitions, Mock Theta Functions, and q-Series**, Online Number Theory Conference and School, July 18-22, 2022 (Online).
- 5/2023 **35th Automorphic Forms Workshop**, LSU, Louisiana, USA, May 22-26, 2023.
- 7/2023 **32èmes Journées Arithmétiques**, Université de Lorraine, France, July 3-7, 2023.
- 6/2024 **Canadian Number Theory Association Meeting**, Fields Institute, Toronto, Canada, June 10-14, 2024.

9/2024 ***Building Bridges: 6th EU/US Summer School & Workshop on Automorphic Forms and Related Topics (BB6) School and Workshop on Automorphic Forms and Related Topics***, CIRM, France, Sep 2-6, 2024.

6/2026 ***Sixth International Workshop on Zeta Functions in Algebra and Geometry***, University of Galway, Galway, Ireland, June 8-12, 2026.

Teaching

9/2025 - **UCC**: Lecturer

MA2013 (Mathematics for Engineering)

MA2200 (Introductory Mathematics for Business II)

MA2054 (Ordinary Differential Equations)

9/2019 - **UCD**: Tutor

Math 10030 (Maths for Business)

Math 10200 (Matrix Algebra)

Math 10210 (Foundations of Mathematics for Computer Science I)

Math 10350 (Calculus in the Mathematical and Physical Sciences)

Math 20320 (Quantitative Methods for Business)

Math 20030 (Linear Algebra 2 for the Mathematical Sciences)

Math 20060 (Calculus of Several Variables)

ACM 10060 (Applications of Differential Equations)

ACM 30030 (Multivariable Calculus for Engineering II)

MST 30070 (Differential Geometry)